STPM 2020 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ *J s,* $c = 3.00 \times 10^{8}$ *m* s^{-1}, $e = 1.60 \times 10^{-19}$ *C.*

Section A [15 marks]

1. A body undergoes a simple harmonic motion. What is the ratio of the potential energy to the total energy when the displacement of the body is $\frac{1}{3}$ of the maximum displacement?

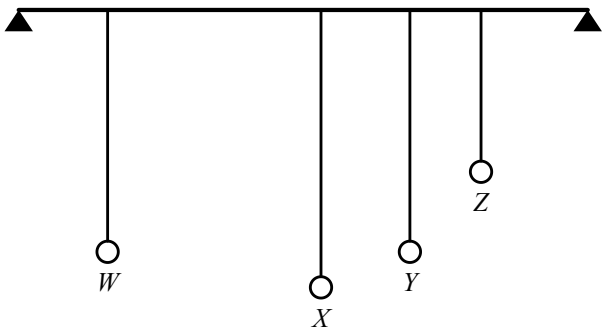
A $\frac{1}{9}$ B $\frac{1}{3}$ C 3 D 9

Answer: A

$$\frac{U}{E} = \frac{½kx^2}{½kA^2} = (\frac{1}{3})^2 = \frac{1}{9}$$

Explanation: Potential energy is proportional to the square of the displacement.

2. Three pendulums with identical light bobs, *X*, *Y* and *Z*, but of different lengths are hung along a horizontal thin wire together with another pendulum which has a heavier bob, *W*, as shown in the diagram below.



W is set into oscillation which forces *X*, *Y* and *Z* to oscillate accordingly. Which statement is true about the forced oscillations?

A *X*, *Y* and *Z* oscillate with the same amplitude as of *W*.
B *X*, *Y* and *Z* oscillate with the same frequency as of *W*.
C *X* oscillates with a lower frequency than *W*.
D *Z* oscillates with a higher frequency than *W*.

Answer: B

In forced oscillation every driven pendulum oscillates at the frequency of the driver *W*.

Explanation: Their amplitudes differ; the pendulum whose natural frequency is closest to that of *W* (same length, *X*) has the largest amplitude (resonance).

3. Which is **not** true about the property of a standing wave?

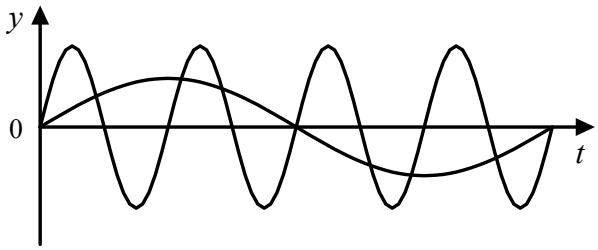
A The net energy transfer is zero.
B The wave profile does not move.
C All particles vibrate with the same amplitude.
D The wave is produced from the superposition of two similar waves which move in the opposite direction.

Answer: C

In a standing wave the amplitude varies with position: zero at nodes, maximum at antinodes. Statement C is not true.

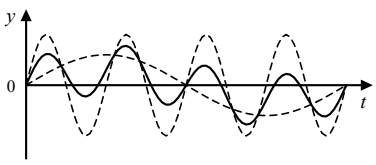
Explanation: A standing wave is formed by two identical waves travelling in opposite directions; it transfers no net energy and its profile does not move.

4. The variations of displacement, y, with time, t, of two progressive waves are shown in the graph below.

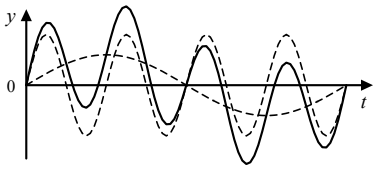


If the two waves approach each other and undergo superposition, which graph shows the correct resultant wave profile?

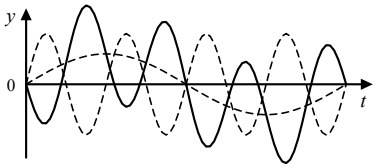
A



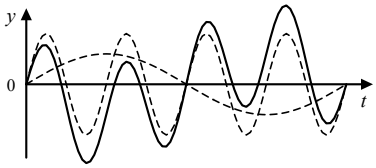
B



C



D



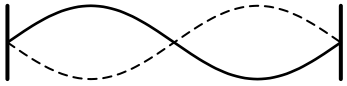
Answer: B

By the principle of superposition, the resultant displacement at each instant is the sum of the two displacements: the large-amplitude, high-frequency wave rides on the low-frequency wave, as in B.

Explanation: Add the two graphs point by point: where the slow wave is positive the fast wave is shifted upward, and vice versa.

5. Which standing wave diagram shows the second overtone of a stretched string?

A



B



C



D



Answer: B

Second overtone = third harmonic: three loops (four nodes) between the fixed ends — diagram B.

Explanation: The fundamental has one loop and the first overtone two loops.

6. A boy cycles towards a stationary source of sound of frequency f and wavelength λ. Which combination is true about the apparent frequency, f', and apparent wavelength, λ', of the sound heard by the boy?

	f'	λ'
A	$f' = f$	$\lambda' = \lambda$
B	$f' < f$	$\lambda' > \lambda$
C	$f' > f$	$\lambda' < \lambda$
D	$f' > f$	$\lambda' = \lambda$

Answer: D

A moving observer meets more wavefronts per second, so $f' > f$; the source is stationary, so the wavelength in the air is unchanged: $\lambda' = \lambda$.

Explanation: Only a moving source changes the wavelength.

7. A man stands 2.00 m from a vertical convex mirror and found that his image is half of his height. What is the focal length of the mirror?

A 0.667 m B 1.33 m C 2.00 m D 4.00 m

Answer: C

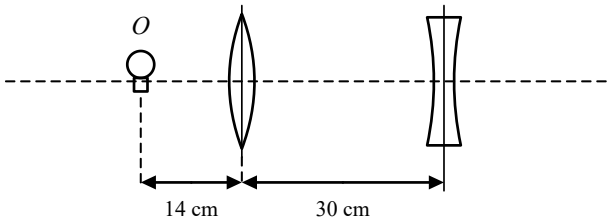
$$m = \frac{1}{2} \Longrightarrow v = -1.00 \text{ m (virtual)}$$

$$\frac{1}{f} = \frac{1}{2.00} + \frac{1}{-1.00} = -0.50 \Longrightarrow f = -2.00 \text{ m}$$

Focal length 2.00 m (convex mirror).

Explanation: A convex mirror always gives a virtual, upright, diminished image behind the mirror.

8. A bright object, O, is placed 14 cm from a converging lens, which is 30 cm to the left of a diverging lens as shown in the diagram below.



If the focal length of the converging lens and the diverging lens is 10 cm and 6.0 cm respectively, what is the distance of the final image from O after refraction from both lenses?

A 14 cm B 44 cm C 60 cm D 74 cm

Answer: D

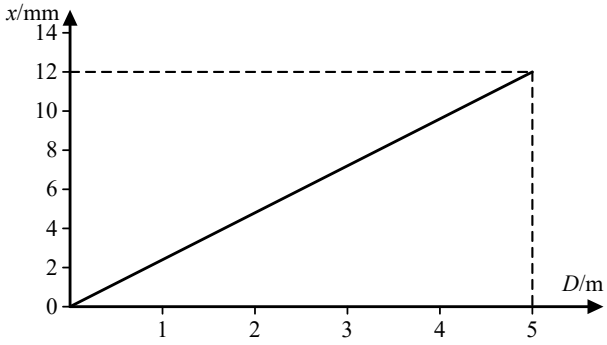
Converging lens: $\frac{1}{v_1} = \frac{1}{10} - \frac{1}{14} \Rightarrow v_1 = 35$ cm, i.e. 5 cm beyond the diverging lens

Diverging lens, virtual object $u_2 = -5.0$ cm: $\frac{1}{v_2} = \frac{1}{-6.0} + \frac{1}{5.0} \Rightarrow v_2 = 30$ cm

Distance from $O = 14 + 30 + 30 = 74$ cm

Explanation: The first image lies beyond the second lens, so it acts as a virtual object for the diverging lens.

9. The variation of fringe separation, x, with the distance of the screen from the double slit, D, using a certain light in a Young's double slit experiment is shown in the graph below.



The wavelengths of the visible light ranges from 400 nm to 700 nm. What is the colour of the fringe formed if the distance between the two slits is 222 μm?

A Green B Orange C Red D Violet

Answer: A

$$\text{Gradient} = \frac{x}{D} = \frac{12 \times 10^{-3}}{5} = 2.4 \times 10^{-3} = \frac{\lambda}{d}$$

$$\lambda = (2.4 \times 10^{-3})(222 \times 10^{-6}) = 5.3 \times 10^{-7} \text{ m} = 533 \text{ nm: green}$$

Explanation: From $x = \lambda D/d$, the gradient of the x–D graph is λ/d.

10. If the number of slits of a diffraction grating is increased but the distance between the slits remains unchanged, the lines in the diffraction pattern will

A move apart and become dimmer
B move closer and become brighter
C become broader and dimmer at the same position
D become narrower and brighter at the same position

Answer: D
The positions of the lines depend only on the slit spacing, so they stay at the same angles; with more slits the lines become sharper (narrower) and brighter.

Explanation: More slits means more waves interfere, giving sharper maxima and more light.

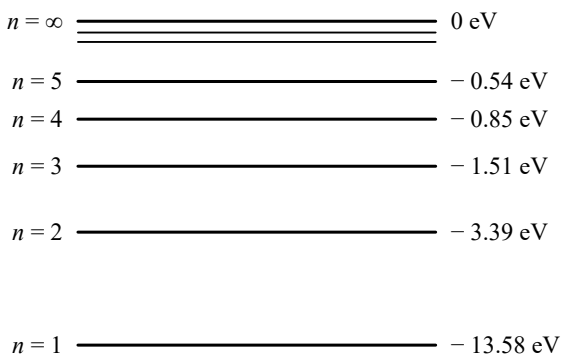
11. In an experiment of a photoelectric effect, the photoelectric current is directly proportional to the intensity of the incident electromagnetic radiation. Which statement is true about the photoelectric current?

A Higher amplitude of the incident radiation causes the emission of more photoelectrons.
B Higher amplitude of the incident radiation causes the emission of faster photoelectrons.
C The larger number of photons per second incident on the photocell causes the emission of photoelectrons at a higher rate.
D The larger number of photons per second incident on the photocell causes the emission of photoelectrons of higher kinetic energy.

Answer: C
A higher intensity means more photons per second, so photoelectrons are emitted at a higher rate, giving a larger current.

Explanation: The kinetic energy of the photoelectrons depends on the photon energy (frequency), not on the number of photons.

12. The energy levels of hydrogen atom are shown in the diagram below.



What is the wavelength of the electromagnetic radiation emitted when the transition of electron from the second excited state to the ground state occurs?

A 103 nm B 122 nm C 367 nm D 823 nm

Answer: A

Second excited state $n = 3 \rightarrow$ ground state: $\Delta E = -1.51 - (-13.58) = 12.07$ eV

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(12.07)(1.60 \times 10^{-19})} = 103 \text{ nm}$$

Explanation: The first excited state is $n = 2$ and the second excited state is $n = 3$.

13. Which is true about nanoscience?

A The study of structures and materials at the nanoscale.
B The research area of nanotechnology that uses methods of quantum mechanics.
C Smaller particles have different magnetic properties compared to larger particles.
D As particles get smaller, the particles tend to react differently with the environment.

Answer: A

Nanoscience is the study of structures and materials on the nanoscale (about 1–100 nm).

Explanation: This is the definition; the other statements describe applications or particular properties.

14. A radioactive sample has an activity of 5.0×10^{5} Bq at a particular instant. The sample decays with a decay constant of 3.9×10^{-12} s^{-1}. What is the number of nuclei in the sample which has not decayed at that instant?

A 7.8×10^{-18} B 2.0×10^{-6} C 5.1×10^{5} D 1.3×10^{17}

Answer: D

$$A = \lambda N \Longrightarrow N = \frac{5.0 \times 10^{5}}{3.9 \times 10^{-12}} = 1.3 \times 10^{17}$$

Explanation: A very small decay constant (long half-life) needs a very large number of nuclei to give a measurable activity.

15. In a nuclear fission, a heavy nucleus splits into two smaller nuclei of smaller masses with emission of several neutrons. Which statement is true about both of the nuclei produced?

A The total nucleon number of the nuclei produced is lower than that of the original nucleus.
B The total nucleon number of the nuclei produced is greater than that of the original nucleus.
C The binding energy per nucleon of the nuclei produced is higher than that of the original nucleus.
D The binding energy per nucleon of the nuclei produced is lower than that of the original nucleus.

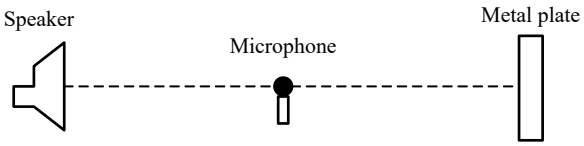
Answer: C

The fission products have a higher binding energy per nucleon than the original heavy nucleus; this is why energy is released.

Explanation: The total nucleon number is conserved (including the emitted neutrons).

Section B [15 marks]

16. In an experiment to determine the speed of sound in air, a speaker, a microphone and a metal plate are arranged in a straight line as shown in the diagram below.



The speaker emits sound of frequency 500 Hz. When the microphone is moved along the line between the speaker and the metal plate, maximum and minimum intensities are detected alternately. The distance between the successive minimum intensities is 34.0 cm.

(a) Explain the formation of the maximum and minimum intensities. [3]

Answer

Sound from the speaker is reflected by the metal plate	1
The incident and reflected waves (same frequency, opposite directions) superpose to form a standing wave	1
Maxima occur at antinodes (constructive interference) and minima at nodes (destructive interference)	1
	3

Explanation: The microphone detects the variation in amplitude along the standing wave.

(b) Determine the speed of sound in air. [3]

Answer

Successive minima (nodes) are $\frac{\lambda}{2}$ apart: $\frac{\lambda}{2} = 0.340$ m	1
$\lambda = 0.680$ m	1
$v = f\lambda = (500)(0.680) = 340$ m s^{-1}	1
	3

Explanation: Measuring across several nodes and dividing reduces the error.

(c) Explain why the minimum intensity is detected at the point nearest to the metal plate. [2]

Answer

The air molecules next to the rigid plate cannot move: there is a displacement node at the plate — 1

(the reflected wave undergoes a phase change of π), so the point nearest the plate is a minimum — 1

2

Total for question **8**

Explanation: A rigid boundary is always a displacement node for a sound wave.

17. A car with the headlamps separated by 1.0 m emits yellow light of wavelength 570 nm approaches a man who is looking at the headlamps. The diameter of the pupils of the man's eyes is 3.0 mm.

(a) Determine the resolving power of the man's eyes. [2]

Answer

$\theta_{\text{min}} = \frac{1.22\lambda}{d} = \frac{1.22(570 \times 10^{-9})}{3.0 \times 10^{-3}}$ — 1

$\theta_{\text{min}} = 2.32 \times 10^{-4}$ rad — 1

2

Explanation: Rayleigh's criterion for a circular aperture (the pupil).

(b) Determine the maximum distance between the man and the car when the yellow light can be resolved. [2]

Answer

$\theta \approx \frac{s}{L} \Rightarrow L = \frac{1.0}{2.32 \times 10^{-4}}$ — 1

$L = 4.31 \times 10^{3}$ m — 1

2

Explanation: Beyond this distance the two headlamps subtend an angle smaller than the limit of resolution and appear as one.

(c) Explain the effect on the maximum distance when the yellow light can be resolved if blue light is emitted. [3]

Answer

Blue light has a shorter wavelength than yellow light	1
so the minimum angle of resolution $1.22\lambda/d$ is smaller (better resolution)	1
and the maximum distance at which the headlamps can be resolved increases	1
	3
Total for question	**7**

Explanation: $L_{max} = sd/1.22\lambda$ is inversely proportional to the wavelength.

Section C [30 marks]

18. Progressive waves are classified into mechanical waves and electromagnetic waves. The progressive wave is represented by
$y = A \sin[(2.8 \times 10^4)(t) - (9.4 \times 10^2)(x)]$,
where x and y are in metres and t in seconds. The amplitude, A, of the wave decreases by 10% after the wave propagates through a distance, r.

(a) State **two** differences and **two** similarities between electromagnetic waves and mechanical waves. [4]

Answer

Difference: electromagnetic waves need no medium; mechanical waves need a medium	1
Difference: all electromagnetic waves travel at 3.00×10^8 m s^{-1} in a vacuum and are transverse; mechanical waves are much slower and may be longitudinal	1
Similarity: both transfer energy	1
Similarity: both show reflection, refraction, interference and diffraction (and obey $v = f\lambda$)	1
	4

Explanation: Electromagnetic waves are oscillating electric and magnetic fields; mechanical waves are vibrations of particles of a medium.

(b) Determine whether the wave is an electromagnetic wave or a mechanical wave. [6]

Answer

Compare with $y = A \sin(\omega t - kx)$: $\omega = 2.8 \times 10^4$ rad s^{-1} 1

$k = 9.4 \times 10^2$ m^{-1} 1

$v = \frac{\omega}{k}$ 1

$v = \frac{2.8 \times 10^4}{9.4 \times 10^2} = 29.8$ m s^{-1} 1

This is far less than the speed of electromagnetic waves (3.00×10^8 m s^{-1}) 1

so it is a mechanical wave 1

6

Explanation: Electromagnetic waves travel at nearly 3×10^8 m s^{-1} in any ordinary medium, so a speed of about 30 m s^{-1} must be mechanical.

(c) Determine the intensity of the wave at r in terms of initial intensity, I_0. [3]

Answer

$I \propto A^2$ 1

New amplitude = $0.90A \Longrightarrow I = (0.90)^2 I_0$ 1

$I = 0.81 I_0$ 1

3

Explanation: A 10% fall in amplitude means a 19% fall in intensity.

(d) State **two** possible reasons why the amplitude decreases when the wave propagates at further distances. [2]

Answer

Energy is absorbed by the medium (dissipated as heat / damping) 1

The energy spreads out over a larger area as the wave travels further from the source 1

2

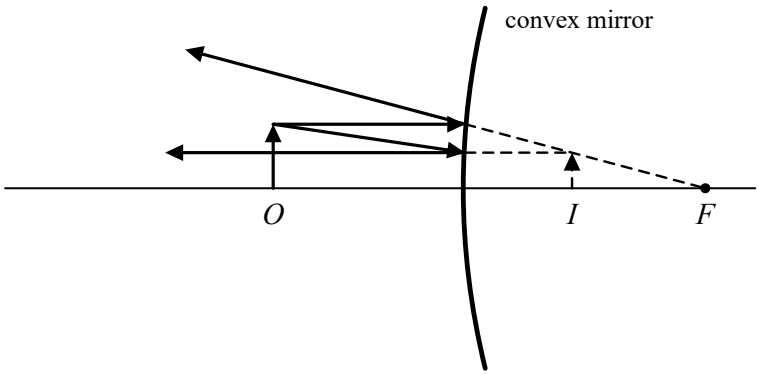
Total for question **15**

Explanation: Both effects reduce the energy per unit area reaching a point.

19. (a) An object of height 4.0 cm is placed on the principal axis, 12.0 cm in front of a convex mirror with a radius of curvature 30.0 cm.

(i) Sketch a ray diagram to show the formation of image. Hence, state **two** characteristics of the image formed. [4]

Answer



Ray parallel to the axis reflected as if coming from F	1
Ray towards F reflected parallel to the axis; extensions meet behind the mirror	1
Characteristics: virtual and upright	1
diminished (between the mirror and F)	1
	4

Explanation: For a convex mirror the reflected rays diverge, so the image is formed where their backward extensions meet.

(ii) Calculate the height of the image. [4]

Answer

$f = -\frac{30.0}{2} = -15.0$ cm	1
$\frac{1}{v} = \frac{1}{-15.0} - \frac{1}{12.0} \Rightarrow v = -6.67$ cm	1
$m = \frac{6.67}{12.0} = 0.556$	1
Image height = (0.556)(4.0) = 2.22 cm	1
	4

Explanation: Use a negative focal length for a convex mirror in the real-is-positive convention.

(b) The convex mirror in (a) is then replaced with a thin biconvex lens with radii of curvature 18.0 cm and refractive index 1.60. A new image is formed.

(i) Determine the ratio of the height of the image to that of (a)(ii). [4]

Answer

$\frac{1}{f} = (1.60 - 1)(\frac{1}{18.0} + \frac{1}{18.0}) \Longrightarrow f = 15.0$ cm — 1

$\frac{1}{v} = \frac{1}{15.0} - \frac{1}{12.0} \Longrightarrow v = -60.0$ cm (virtual) — 1

$m = \frac{60.0}{12.0} = 5.0 \Longrightarrow$ image height = 20.0 cm — 1

Ratio = $\frac{20.0}{2.22} = 9.0$ — 1

4

Explanation: The object is inside the focal length of the converging lens, so the lens acts as a magnifying glass.

(ii) Explain the results of the experiment if the whole set up of the experiment is placed in water. [3]

Answer

In water the refractive index of the lens relative to water falls to 1.60/1.33 = 1.20, so the focal length of the lens increases (to about 44 cm) — 1

The lens image is still virtual and upright but nearer and less magnified (about 1.4 times) — 1

The mirror's focal length does not change (reflection does not depend on the medium), so its image is unchanged — 1

3

Total for question **15**

Explanation: The lensmaker's equation uses the refractive index of the lens relative to its surroundings.

20. (a) Describe the production of a characteristic spectrum of X-rays. [3]

Answer

High-energy electrons hitting the target knock out electrons from inner shells (e.g. the K shell) of the target atoms — 1

Electrons from higher shells fall into the vacancies — 1

emitting X-ray photons of definite energies (wavelengths) characteristic of the target element, e.g. K_α and K_β lines — 1

3

Explanation: The photon energy equals the difference between the two energy levels involved.

(b)

(i) Copper is used as the target material of anode. The potential difference applied to the anode and cathode is 23.0 kV. Calculate the minimum wavelength of the X-rays produced. [4]

Answer

$eV = \frac{hc}{\lambda_{min}}$ 1

$\lambda_{min} = \frac{hc}{eV}$ 1

$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(1.60 \times 10^{-19})(23.0 \times 10^{3})}$ 1

$= 5.40 \times 10^{-11}$ m 1

4

Explanation: The minimum wavelength does not depend on the target material.

(ii) State the effect on minimum wavelength if the potential difference is increased. [1]

Answer

The minimum wavelength decreases 1

1

Explanation: $\lambda_{min} \propto 1/V$.

(c) When X-rays is incident on atomic planes, the waves are reflected and a diffraction pattern is produced. The first order of the diffraction is observed at glancing angle 26.2° from a crystal with spacing between the planes is 0.270 nm.

(i) Determine the wavelength of the X-ray used. [2]

Answer

$2d \sin\theta = n\lambda \Longrightarrow \lambda = 2(0.270 \times 10^{-9}) \sin 26.2°$ 1

$\lambda = 2.38 \times 10^{-10}$ m 1

2

Explanation: First order: $n = 1$. The glancing angle is measured from the atomic planes.

(ii) Determine the maximum order of constructive interference that could be observed. [2]

Answer

$\sin\theta \leq 1 \Longrightarrow n \leq \frac{2d}{\lambda} = \frac{2(0.270 \times 10^{-9})}{2.38 \times 10^{-10}} = 2.27$ 1

Maximum order $n = 2$ 1

2

Explanation: The order must be a whole number.

(iii) Describe X-ray diffraction by two parallel adjacent atomic planes. [3]

Answer

X-rays are reflected (scattered) by atoms in each of the two parallel planes	1
The ray reflected from the lower plane travels an extra distance $2d \sin \theta$	1
Constructive interference (a strong diffracted beam) occurs when $2d \sin \theta = n\lambda$ (Bragg's law)	1
	3
Total for question	**15**

Explanation: At other angles the reflections from many planes cancel out.